

Kinora Lower Leg

Clinical AI Orientation (Part 9)

Structured reference for RAG / AI-assisted physiotherapy consultation

Bones, Compartments, Muscles, Nerves, Vessels, Fascia, Biomechanics, Pathologies &
Rehabilitation

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Disclaimer

*This document is an educational orientation resource for Kinora AI clinical consultation support. It is NOT a substitute for professional clinical judgment, direct patient examination, or licensed care. Content reflects Gray's/Standring, Moore, Netter, Neumann, Magee, Brukner & Khan, and sports medicine themes for MTSS, bone stress injury, CECS, and calf injury. **Red flags — acute compartment syndrome, suspected DVT/PE, Achilles rupture, high-risk/open fracture, or acute foot drop with trauma — require urgent medical/orthopaedic referral.** Acute compartment syndrome is a surgical emergency; do not delay for imaging if clinically evident.*

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1. Bones

RECORD: Tibia

ID: LL-BONE-01

Bone: Tibia (shin bone) — primary weight-bearing bone of the leg

Region: Lower leg — medial

Landmarks: Tibial tuberosity; anterior border (shin); medial malleolus; soleal line; nutrient foramen; interosseous border; fibular notch distally; Gerdy's tubercle proximally

Articulations: Proximal: femur (via menisci), proximal fibula. Distal: talus (ankle mortise), distal fibula (syndesmosis)

Muscle Attachments: Pes anserinus, tibialis anterior, EDL, soleus, FDL, tibialis posterior, FHL (more fibular/interosseous), popliteus, gastroc not direct shaft

Ligament Attachments: MCL distal; coronary/meniscotibial proximally; deltoid ligament at medial malleolus; syndesmotic ligaments distally; interosseous membrane along shaft

Blood Supply: Nutrient artery from posterior tibial; periosteal vessels from anterior/posterior tibial; genicular contributions proximally

Innervation: Periosteal branches related to saphenous (medial), deep fibular/tibial regionally — bone pain in stress injury often diffuse

Biomechanical Role: Transmits ~80-100% of axial load in stance (fibula shares less); medial malleolus forms ankle mortise

Clinical Importance: Most common site of tibial stress fracture and MTSS; open fractures high infection risk; anterior border subcutaneous — contusion/fracture palpable

Fractures: Shaft (AO/OTA); plateau (proximal); pilon (distal); stress fracture posteromedial cortex classic in runners

Healing Timeline: Stress injury weeks–months by grade; complete shaft often 3–6+ months; delayed/nonunion risk with poor biology/smoking/NSAIDs overuse

Imaging: XR first for trauma; MRI preferred for stress injury; CT for complex articular; bone scan adjunct historical

Rehabilitation: Unload per grade (boot/crutches); progressive loading guided by pain and imaging stage; address training errors, calf capacity, footwear/surface

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Bone stress injury consensus themes (e.g. ACSM/sports medicine reviews).

RECORD: Fibula

ID: LL-BONE-02

Bone: Fibula — lateral non-primary axial load bone of the leg

Region: Lower leg — lateral

Landmarks: Fibular head; neck (common fibular nerve); shaft; lateral malleolus; malleolar fossa

Articulations: Proximal tibiofibular joint; distal tibiofibular syndesmosis; talus laterally (ankle)

Muscle Attachments: Biceps femoris (head); soleus; fibularis longus/brevis; FHL; tibialis posterior (via interosseous); EDL/EHL near interosseous

Ligament Attachments: LCL and PLC at head; interosseous membrane; AITFL/PITFL/inferior transverse at syndesmosis; lateral ankle ligaments at malleolus

Blood Supply: Fibular (peroneal) artery nutrient/periosteal branches; genicular proximally

Innervation: Common fibular nerve closely related at neck — critical clinical landmark

Biomechanical Role: Lateral ankle stability; syndesmosis integrity; ~6-30% load share cited variously; muscle attachment strut

Clinical Importance: Fibular neck fracture → foot drop risk; Maisonneuve fracture with ankle injury; stress fractures in runners/dancers; lateral malleolus fractures

Fractures: Head/neck; shaft; distal (Weber/Lauge-Hansen ankle classifications); stress fracture

Healing Timeline: Often 6-10 weeks isolated; dictated by ankle/syndesmosis involvement

Imaging: Full-length tibia-fibula views if Maisonneuve suspected; MRI for stress; CT syndesmosis

Rehabilitation: Protect nerve; progressive eversion strength; ankle stability; WB per fracture pattern

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

2. Compartments

RECORD: Anterior Compartment

ID: LL-COMP-01

Compartment: Anterior compartment of the leg

Boundaries: Tibia, interosseous membrane, anterior intermuscular septum, crural fascia

Muscles: Tibialis anterior, EHL, EDL, fibularis tertius

Nerves: Deep fibular nerve

Arteries: Anterior tibial artery

Veins: Anterior tibial veins (venae comitantes)

Function: Ankle dorsiflexion; toe extension; foot inversion (TA) / eversion assist (tertus)

Biomechanics: Eccentric control of plantarflexion at heel strike; clears foot in swing

Clinical Importance: Foot drop if deep fibular injured; common CECS site; shin contusion vulnerability

Compartment Syndrome: Acute: pain out of proportion, pain on passive stretch (PF), paresthesia 1st web — emergency fasciotomy. Chronic exertional: reversible pressure with exercise

Assessment: Passive stretch pain, neurovascular exam, needle manometry for CECS protocols, MRI/SPECT selected

Treatment: Acute = urgent fasciotomy. CECS = activity modification, gait retraining, fasciotomy if refractory

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Acute compartment syndrome orthopaedic emergencies literature.

RECORD: Lateral Compartment

ID: LL-COMP-02

Compartment: Lateral compartment of the leg

Boundaries: Fibula, anterior and posterior intermuscular septa, crural fascia

Muscles: Fibularis (peroneus) longus and brevis

Nerves: Superficial fibular nerve

Arteries: Branches from fibular artery / perforators (no major named artery exclusively inside)

Veins: Accompanying veins / perforators

Function: Foot eversion; PL plantarflexes 1st ray; ankle lateral dynamic stability

Biomechanics: Control inversion moments; important on uneven ground and cutting

Clinical Importance: Peroneal tendinopathy/subluxation; superficial fibular sensory entrapment; CECS less common than anterior/deep posterior

Compartment Syndrome: Possible but less frequent; assess with lateral pain and passive inversion stretch pain

Assessment: Resisted eversion, tendon palpation behind lateral malleolus, sensory map dorsum foot

Treatment: Tendinopathy loading; acute CS fasciotomy; address ankle instability drivers

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Superficial Posterior Compartment

ID: LL-COMP-03

Compartment: Superficial posterior compartment

Boundaries: Transverse intermuscular septum deep; crural fascia superficial; tibia/fibula margins via septa

Muscles: Gastrocnemius, soleus, plantaris — triceps surae forming Achilles

Nerves: Tibial nerve branches (motor)

Arteries: Posterior tibial and sural branches of popliteal

Veins: Small saphenous / sural veins related superficially; deep venae comitantes

Function: Powerful plantarflexion; soleus postural; gastroc also knee flexion

Biomechanics: Propulsion in gait; soleus high stance EMG; Achilles energy storage in running

Clinical Importance: Achilles tendinopathy continuum; tennis leg (medial gastroc); DVT differential for calf pain

Compartment Syndrome: Less common isolated; deep posterior more classic for CECS/MTSS overlap region

Assessment: Thompson test for Achilles continuity; calf squeeze; vascular screen if acute swelling

Treatment: Progressive loading for tendinopathy/strain; emergency care if acute CS/DVT suspected

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Achilles tendinopathy loading literature.

RECORD: Deep Posterior Compartment

ID: LL-COMP-04

Compartment: Deep posterior compartment

Boundaries: Tibia, fibula, interosseous membrane, transverse intermuscular septum

Muscles: Tibialis posterior, FDL, FHL; popliteus proximally related

Nerves: Tibial nerve

Arteries: Posterior tibial artery; fibular artery courses in/near this region

Veins: Posterior tibial and fibular venae comitantes

Function: Plantarflexion, inversion (TP), toe flexion; supports medial longitudinal arch (TP)

Biomechanics: TP eccentrically controls pronation in stance; FHL/FDL toe control

Clinical Importance: CECS common here; TP dysfunction/AHD continuum; tarsal tunnel distal tibial nerve

Compartment Syndrome: Acute and chronic — pain on passive eversion/dorsiflexion of toes; emergency if acute

Assessment: Toe flexion strength, TP single-heel-rise inversion, CECS testing protocols

Treatment: Acute fasciotomy; CECS conservative vs fasciotomy; TP progressive loading

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. CECS reviews.

3. Muscles

RECORD: Tibialis Anterior

ID: LL-MUS-01

Muscle: Tibialis anterior

Origin: Lateral tibial condyle and upper lateral tibia + IO membrane

Insertion: Medial cuneiform and base of 1st MT

Innervation: Deep fibular nerve (L4-L5)

Blood Supply: Anterior tibial artery

Fiber Direction: Anterolateral tibia vertical

Primary Action: Ankle dorsiflexion and inversion

Secondary Action: Supports medial arch dynamically

Stabilizing Function: Eccentric control of foot slap at heel strike

Synergists: EHL, EDL

Antagonists: Gastroc/soleus, fibularis longus

Length-Tension Relationship: Active insufficiency less relevant; high demand eccentric heel strike

EMG Activation: High early stance eccentric and swing concentric

Trigger Points: Anterior shin

Pain Referral: Anterior ankle/shin

Stretch: Plantarflexion-eversion stretch

Strengthening: Heel walks, resisted DF, eccentric lowering

Neuromuscular Control: Heel-strike control; prevent foot slap

Clinical Importance: Foot drop; anterior CECS; MTSS differential

Common Injuries: Strain; contusion over subcutaneous tibia

Rehabilitation: Progressive DF loading; gait cadence; address footwear

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Extensor Hallucis Longus

ID: LL-MUS-02

Muscle: Extensor hallucis longus

Origin: Middle fibula and IO membrane

Insertion: Dorsal base distal phalanx hallux

Innervation: Deep fibular nerve (L5)

Blood Supply: Anterior tibial artery

Fiber Direction: Deep anterior compartment

Primary Action: Extends hallux; assists dorsiflexion

Secondary Action: Assists inversion slightly

Stabilizing Function: Toe clearance in swing

Synergists: EDL, TA

Antagonists: FHL

Length-Tension Relationship: Isolated hallux extension testing

EMG Activation: Swing phase hallux control

Trigger Points: Dorsal foot/ankle

Pain Referral: 1st ray dorsal

Stretch: Plantarflexion stretch with hallux flexed

Strengthening: Resisted hallux extension; toe yoga drills

Neuromuscular Control: Push-off hallux alignment

Clinical Importance: L5 myotome testing; anterior CS stretch pain

Common Injuries: Strain rare; laceration risk dorsum foot

Rehabilitation: Progressive hallux loading

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Extensor Digitorum Longus

ID: LL-MUS-03

Muscle: Extensor digitorum longus

Origin: Lateral tibial condyle, proximal fibula, IO membrane

Insertion: Middle and distal phalanges toes 2-5 (extensor expansion)

Innervation: Deep fibular nerve (L4-L5)

Blood Supply: Anterior tibial artery

Fiber Direction: Anterior compartment lateral to TA/EHL

Primary Action: Extends toes 2-5; assists dorsiflexion

Secondary Action: Weak eversion

Stabilizing Function: Toe clearance

Synergists: EHL, TA, fibularis tertius

Antagonists: FDL

Length-Tension Relationship: Works with extensor digitorum brevis

EMG Activation: Swing toe extension EMG

Trigger Points: Anterior ankle

Pain Referral: Dorsum toes

Stretch: Plantarflexion stretch

Strengthening: Resisted toe extension; heel walks

Neuromuscular Control: Swing clearance

Clinical Importance: Anterior CS; L5 testing

Common Injuries: Strain uncommon

Rehabilitation: DF endurance training

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Fibularis Tertius

ID: LL-MUS-04

Muscle: Fibularis (peroneus) tertius

Origin: Distal fibula / EDL continuum (variable)

Insertion: Dorsal base 5th MT

Innervation: Deep fibular nerve (L5-S1)

Blood Supply: Anterior tibial artery

Fiber Direction: Anterolateral distal leg — variable presence

Primary Action: Dorsiflexion and eversion

Secondary Action: May tension 5th MT region

Stabilizing Function: Lateral ankle dynamic assist in DF-eversion

Synergists: EDL, fibularis brevis

Antagonists: TA (inversion), plantarflexors

Length-Tension Relationship: Absent in some individuals — anatomic variant

EMG Activation: Low-moderate eversion-DF tasks

Trigger Points: Lateral ankle

Pain Referral: 5th MT base

Stretch: Inversion-PF stretch

Strengthening: Resisted DF-eversion

Neuromuscular Control: Uneven ground eversion control

Clinical Importance: Differentiate from fibularis brevis injury at 5th MT

Common Injuries: Rare isolated injury

Rehabilitation: Ankle stability progressions

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Fibularis Longus

ID: LL-MUS-05

Muscle: Fibularis (peroneus) longus

Origin: Fibular head and proximal lateral fibula

Insertion: Plantar medial cuneiform and 1st MT base (after cuboid groove)

Innervation: Superficial fibular nerve (L5-S1)

Blood Supply: Fibular artery perforators

Fiber Direction: Lateral compartment — longest course under foot

Primary Action: Eversion; plantarflexes 1st ray

Secondary Action: Plantarflexion assist; stabilizes 1st ray against TA

Stabilizing Function: Lateral ankle dynamic stabilizer; supports transverse/lateral arch mechanics

Synergists: Fibularis brevis

Antagonists: TA, TP

Length-Tension Relationship: Tendon turns at lateral malleolus and cuboid — high friction zones

EMG Activation: High on uneven terrain and cutting

Trigger Points: Lateral leg

Pain Referral: Lateral ankle/foot

Stretch: Inversion stretch

Strengthening: Resisted eversion, band walks, single-leg balance on uneven surface

Neuromuscular Control: Cut/cut landing eversion control

Clinical Importance: Tendinopathy; subluxation with retinaculum injury; cuboid syndrome association

Common Injuries: Tendinopathy; tears; superior retinaculum injury

Rehabilitation: Progressive eversion loading; address ankle instability

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Fibularis Brevis

ID: LL-MUS-06

Muscle: Fibularis (peroneus) brevis

Origin: Distal lateral fibula

Insertion: Tuberosity of 5th MT base

Innervation: Superficial fibular nerve (L5-S1)

Blood Supply: Fibular artery perforators

Fiber Direction: Lateral compartment deep to longus

Primary Action: Eversion; weak plantarflexion

Secondary Action: Lateral ankle stability

Stabilizing Function: Primary everter at malleolus level

Synergists: Fibularis longus

Antagonists: TA, TP

Length-Tension Relationship: Insertion avulsion / Jones fracture differential at 5th MT

EMG Activation: Eversion tasks high EMG

Trigger Points: Lateral ankle

Pain Referral: 5th MT base

Stretch: Inversion stretch

Strengthening: Resisted eversion; proprioceptive ankle drills

Neuromuscular Control: Landing eversion control

Clinical Importance: Most common peroneal tendon tear; 5th MT avulsion

Common Injuries: Longitudinal split tears; tendinopathy

Rehabilitation: Load progressive; surgical if unstable subluxation/tear selected

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Gastrocnemius

ID: LL-MUS-07

Muscle: Gastrocnemius

Origin: Medial and lateral femoral condyles

Insertion: Achilles tendon to calcaneus (with soleus)

Innervation: Tibial nerve (S1-S2)

Blood Supply: Sural branches of popliteal

Fiber Direction: Biarticular posterior — crosses knee and ankle

Primary Action: Powerful plantarflexion; knee flexion assist

Secondary Action: Propulsion in sprint/jump

Stabilizing Function: Dynamic posterior knee restraint to hyperextension

Synergists: Soleus, plantaris

Antagonists: TA, tibialis anterior group

Length-Tension Relationship: Active insufficiency knee flexed + PF; medial head strain classic (tennis leg)

EMG Activation: High sprint push-off and jump EMG

Trigger Points: Calf

Pain Referral: Posterior knee/calf — DVT differential critical

Stretch: Wall stretch knee extended

Strengthening: Heel raises knee extended; plyometrics graded

Neuromuscular Control: Stiffness modulation landing

Clinical Importance: Tennis leg; contributes Achilles load

Common Injuries: Medial head myofascial tear

Rehabilitation: Progressive loading; rule out DVT if acute swelling

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Soleus

ID: LL-MUS-08

Muscle: Soleus

Origin: Soleal line of tibia, proximal fibula, IO arch

Insertion: Achilles tendon to calcaneus

Innervation: Tibial nerve (S1-S2)

Blood Supply: Posterior tibial, fibular, popliteal branches

Fiber Direction: Deep to gastroc — monoarticular ankle

Primary Action: Plantarflexion (primary postural)

Secondary Action: Venous pump of leg

Stabilizing Function: Controls tibial advancement in midstance (eccentric)

Synergists: Gastroc, TP, FHL

Antagonists: TA

Length-Tension Relationship: Best strengthened with knee flexed heel raises

EMG Activation: High continuous stance EMG

Trigger Points: Deep calf

Pain Referral: Achilles/heel — soleus strain often deep pain

Stretch: Knee-flexed calf stretch

Strengthening: Seated/bent-knee heel raises, heavy slow Achilles loading

Neuromuscular Control: Midstance control; cadence

Clinical Importance: Achilles tendinopathy; soleus strain misdiagnosed as DVT/MTSS

Common Injuries: Strain; contusion

Rehabilitation: Heavy slow resistance; graded return to run

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Plantaris

ID: LL-MUS-09

Muscle: Plantaris

Origin: Lateral supracondylar femur / oblique popliteal ligament region

Insertion: Medial Achilles / calcaneus (variable)

Innervation: Tibial nerve (S1-S2)

Blood Supply: Sural/popliteal branches

Fiber Direction: Tiny muscle long tendon — vestigial-like

Primary Action: Weak plantarflexion and knee flexion

Secondary Action: Proprioceptive role proposed

Stabilizing Function: Minimal torque — tendon used as graft sometimes

Synergists: Gastroc

Antagonists: TA

Length-Tension Relationship: Tendon rupture may cause sharp calf pain with intact Achilles (Thompson negative)

EMG Activation: Low EMG contribution

Trigger Points: Calf

Pain Referral: Posteromedial calf

Stretch: With gastroc stretch

Strengthening: Not primary strengthening target

Neuromuscular Control: Differentiate from tennis leg/Achilles rupture

Clinical Importance: Plantaris rupture; graft harvest

Common Injuries: Tendon rupture

Rehabilitation: Symptom-guided; reassure if Achilles intact

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Tibialis Posterior

ID: LL-MUS-10

Muscle: Tibialis posterior

Origin: IO membrane, posterior tibia and fibula

Insertion: Navicular tuberosity and plantar tarsals/MTs (broad)

Innervation: Tibial nerve (L4-L5)

Blood Supply: Posterior tibial artery

Fiber Direction: Deep posterior — closest to IO membrane

Primary Action: Inversion and plantarflexion; key medial arch supporter

Secondary Action: Controls pronation eccentrically

Stabilizing Function: Dynamic medial longitudinal arch stabilizer

Synergists: FDL, FHL, TA (inversion)

Antagonists: Fibularis longus/brevis

Length-Tension Relationship: Dysfunction linked to adult-acquired flatfoot continuum

EMG Activation: High midstance eccentric in runners

Trigger Points: Medial shin/arch

Pain Referral: Medial ankle/arch

Stretch: Eversion-DF stretch carefully

Strengthening: Inversion strength, short-foot, heel raise with inversion bias, arch control drills

Neuromuscular Control: Single-leg stance arch control

Clinical Importance: PTTD; CECS deep posterior; MTSS differential

Common Injuries: Tendinopathy; rupture in degenerative PTTD

Rehabilitation: Progressive loading; orthoses selected; surgical stages for PTTD

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Flexor Hallucis Longus

ID: LL-MUS-11

Muscle: Flexor hallucis longus

Origin: Distal posterior fibula and IO membrane

Insertion: Distal phalanx hallux plantar

Innervation: Tibial nerve (S2-S3)

Blood Supply: Fibular / posterior tibial branches

Fiber Direction: Deep posterior — lateral

Primary Action: Flexes hallux; assists plantarflexion

Secondary Action: Supports medial arch

Stabilizing Function: Push-off hallux flexion force

Synergists: FDL, TP

Antagonists: EHL

Length-Tension Relationship: Henry's knot intersection with FDL; stenosing tenosynovitis in dancers (trigger toe)

EMG Activation: Push-off high in dance/sprint

Trigger Points: Posteromedial ankle

Pain Referral: Plantar hallux

Stretch: Hallux extension stretch

Strengthening: Toe curls, marble pickups, calf raises with toe flexion

Neuromuscular Control: Push-off mechanics

Clinical Importance: Dancer's tendinopathy; posteromedial ankle impingement overlap

Common Injuries: Tenosynovitis; strain

Rehabilitation: Load management; progressive hallux flexion strength

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Flexor Digitorum Longus

ID: LL-MUS-12

Muscle: Flexor digitorum longus

Origin: Posterior tibia below soleal line

Insertion: Distal phalanges toes 2-5

Innervation: Tibial nerve (S2-S3)

Blood Supply: Posterior tibial artery

Fiber Direction: Deep posterior — medial

Primary Action: Flexes toes 2-5; assists PF and inversion

Secondary Action: Arch support assist

Stabilizing Function: Toe grip in stance

Synergists: FHL, TP

Antagonists: EDL

Length-Tension Relationship: Crosses FHL at Henry knot

EMG Activation: Stance toe flexion EMG

Trigger Points: Medial shin

Pain Referral: Plantar toes

Stretch: Toe extension stretch

Strengthening: Towel curls; intrinsic foot + FDL loading

Neuromuscular Control: Push-off toe flexion

Clinical Importance: Deep posterior CECS; tarsal tunnel differential

Common Injuries: Strain uncommon

Rehabilitation: Progressive toe flexor loading

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Popliteus

ID: LL-MUS-13

Muscle: Popliteus

Origin: Lateral femoral condyle and fibular head via popliteofibular complex

Insertion: Posterior proximal tibia

Innervation: Tibial nerve (L4-S1)

Blood Supply: Popliteal / inferior medial genicular

Fiber Direction: Floor of popliteal fossa — oblique

Primary Action: Unlocks knee (tibial IR open chain)

Secondary Action: Dynamic PLC stabilizer

Stabilizing Function: Rotational control near extension

Synergists: PLC ligaments

Antagonists: Knee extensors / tibial ER

Length-Tension Relationship: Screw-home reversal initiator

EMG Activation: Flexion initiation EMG

Trigger Points: Posterolateral knee

Pain Referral: Posterolateral knee

Stretch: Gentle ER stretch variants

Strengthening: Tibial IR control drills

Neuromuscular Control: PLC rehab

Clinical Importance: PLC injury association

Common Injuries: With PLC trauma

Rehabilitation: Protect varus/ER early post-PLC surgery

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

4. Nerves

RECORD: Tibial Nerve

ID: LL-NRV-01

Nerve: Tibial nerve

Root Levels: L4-S3

Motor Function: Posterior compartment muscles (gastroc, soleus, TP, FDL, FHL, popliteus) then plantar intrinsics via medial/lateral plantar

Sensory Distribution: Posterolateral leg via sural contributions; heel/plantar foot via calcaneal and plantar nerves

Dermatomes: S1-S2 plantar emphasis

Myotomes: S1-S2 plantarflexion / toe flexion

Entrapment Sites: Popliteal fossa; tarsal tunnel behind medial malleolus

Clinical Presentation: Calf weakness, plantar numbness, tarsal tunnel burning pain

Neurodynamic Tests: SLR with DF/eversion tibial bias

Differential Diagnosis: S1 radiculopathy, CECS, Achilles pathology, DVT

EMG: Localizes tibial vs root vs plantar nerve

Imaging: US/MRI tarsal tunnel or popliteal mass

Treatment: Relieve entrapment, orthoses, surgical release selected

Rehabilitation: Nerve mobility, calf/foot strength, load management

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Common Fibular Nerve

ID: LL-NRV-02

Nerve: Common fibular (peroneal) nerve

Root Levels: L4-S2

Motor Function: Short head biceps femoris; then divides to deep and superficial fibular

Sensory Distribution: Lateral leg proximal; continues via deep/superficial branches

Dermatomes: L5 lateral leg/dorsum foot

Myotomes: L4-L5 DF/eversion

Entrapment Sites: Fibular neck — cast, leg crossing, trauma, surgery

Clinical Presentation: Foot drop, steppage gait, sensory loss dorsum foot

Neurodynamic Tests: SLR with PF/inversion fibular bias

Differential Diagnosis: L4-L5 radiculopathy, anterior CS, sciatic injury

EMG: Critical for prognosis/localization

Imaging: US at fibular neck; MRI if mass

Treatment: Relieve compression, AFO, decompression selected

Rehabilitation: AFO gait, NMES DF, skin protection, progressive strength

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Deep Fibular Nerve

ID: LL-NRV-03

Nerve: Deep fibular nerve

Root Levels: L4-L5 (from common fibular)

Motor Function: Anterior compartment — TA, EHL, EDL, fibularis tertius; EDB/EHB in foot

Sensory Distribution: 1st web space dorsal

Dermatomes: L4-L5 web space

Myotomes: L4-L5 dorsiflexion / hallux extension

Entrapment Sites: Anterior compartment pressure; anterior tarsal tunnel under extensor retinaculum

Clinical Presentation: Foot drop, 1st web numbness; anterior ankle pain if entrapment

Neurodynamic Tests: Fibular-biased SLR; local retinacular Tinel

Differential Diagnosis: Common fibular lesion, L4-L5 root, CECS

EMG: Distinguishes deep vs common fibular

Imaging: US/MRI anterior ankle

Treatment: Pressure relief, activity mod for CECS, release selected

Rehabilitation: DF strengthening, gait, CECS return-to-run plans

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Superficial Fibular Nerve

ID: LL-NRV-04

Nerve: Superficial fibular nerve

Root Levels: L5-S1

Motor Function: Fibularis longus and brevis

Sensory Distribution: Most of dorsum of foot (except 1st web and sural lateral border)

Dermatomes: L5 dorsum foot

Myotomes: L5-S1 eversion

Entrapment Sites: Exits crural fascia mid-lateral leg — fascial exit entrapment; ankle inversion trauma stretch

Clinical Presentation: Lateral leg/dorsal foot paresthesia; weak eversion if motor involved

Neurodynamic Tests: Fibular bias; local Tinel at fascial exit

Differential Diagnosis: L5 radiculopathy, peroneal tendinopathy, lateral ankle sprain sequelae

EMG: May show sensory changes

Imaging: US fascial exit

Treatment: Fascial release selected; ankle stability rehab

Rehabilitation: Eversion strength, neurodynamics, graded running

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Sural Nerve

ID: LL-NRV-05

Nerve: Sural nerve

Root Levels: S1-S2 (tibial + common fibular contributions typical)

Motor Function: None (pure sensory)

Sensory Distribution: Posterolateral distal leg and lateral foot border

Dermatomes: S1 lateral foot

Myotomes: N/A

Entrapment Sites: Lateral ankle; Achilles surgery; tight boots; sural nerve biopsy site neuroma

Clinical Presentation: Burning lateral foot/ankle; Tinel along sural course

Neurodynamic Tests: SLR with DF/inversion sural bias

Differential Diagnosis: S1 radiculopathy, lateral ankle ligament pain, Achilles-related

EMG: SNAP useful

Imaging: US neuroma

Treatment: Desensitization, padding, surgical neuroma selected

Rehabilitation: Nerve mobility, graded exposure, ankle strength

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Saphenous Nerve

ID: LL-NRV-06

Nerve: Saphenous nerve (leg)

Root Levels: L3-L4 (femoral continuation)

Motor Function: None

Sensory Distribution: Medial leg to medial malleolus/foot border

Dermatomes: L4 medial leg

Myotomes: N/A

Entrapment Sites: Hunter canal; medial knee surgery; along great saphenous vein harvest

Clinical Presentation: Medial shin/ankle numbness or neuropathic pain — MTSS differential

Neurodynamic Tests: Saphenous neurodynamics

Differential Diagnosis: MTSS, tibial stress fracture, L4 radiculopathy

EMG: Sensory NCS

Imaging: US if neuroma

Treatment: Desensitization, address entrapment

Rehabilitation: Load management of medial shin with neuropathic care if needed

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

5. Blood Supply

RECORD: Anterior Tibial Artery

ID: LL-ART-01

Artery: Anterior tibial artery

Origin: Popliteal artery bifurcation (with posterior tibial)

Course: Through IO membrane to anterior compartment; becomes dorsalis pedis at ankle

Branches: Anterior/posterior tibial recurrent; muscular; malleolar; continues as dorsalis pedis

Structures Supplied: Anterior compartment muscles; anterior tibia periosteum; dorsum foot via DP

Venous Drainage: Anterior tibial veins → popliteal

Clinical Importance: Dorsalis pedis pulse; injury in anterior CS/trauma; vascular exam with fractures

Imaging: Duplex/CTA/angiography trauma

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH.

RECORD: Posterior Tibial Artery

ID: LL-ART-02

Artery: Posterior tibial artery

Origin: Popliteal bifurcation

Course: Deep posterior compartment behind medial malleolus into foot (medial/lateral plantar)

Branches: Circumflex fibular, nutrient tibial, muscular, medial malleolar; gives fibular artery (often)

Structures Supplied: Posterior compartments; tibia; plantar foot

Venous Drainage: Posterior tibial veins → popliteal

Clinical Importance: Posterior tibial pulse; tarsal tunnel companion; critical limb ischemia assessment

Imaging: ABI, duplex, CTA

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH.

RECORD: Fibular Artery

ID: LL-ART-03

Artery: Fibular (peroneal) artery

Origin: Usually posterior tibial artery (proximal)

Course: Along fibula in deep posterior / flexor hallucis plane; perforators to lateral compartment

Branches: Nutrient fibular; perforating; communicating; malleolar branches

Structures Supplied: Fibula; deep posterior lateral muscles; lateral compartment via perforators; ankle anastomoses

Venous Drainage: Fibular veins → posterior tibial / popliteal

Clinical Importance: Fibular graft vascular pedicle; lateral ankle trauma; collateral pathway if PTA diseased

Imaging: CTA/angiography for flap/trauma planning

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH.

6. Fascia

RECORD: Crural Fascia

ID: LL-FAS-01

Structure: Crural (deep) fascia of the leg

Attachments: Continuous with fascia lata above; attaches to tibial periosteum medially (subcutaneous tibia); forms retinacula at ankle

Continuity: Blends with periosteum over medial tibia; continuous with extensor/flexor/fibular retinacula; septa inward to fibula/tibia

Force Transmission: Contains muscle expansion; contributes to venous pump; transmits tension from muscle contraction to bone (traction periostitis concept in MTSS)

Clinical Importance: Forms compartment walls; MTSS traction hypothesis along medial tibial fascia; CECS noncompliance of fascia

Restrictions: Thick/inelastic fascia implicated in CECS; post-traumatic scarring

Assessment: Palpation medial tibial border; compartment firmness post-exercise; fascial exit points of superficial fibular nerve

Treatment: Load management; gait retraining; fasciotomy for refractory CECS; soft-tissue techniques adjunctive (limited evidence alone)

Evidence: Fasciotomy outcomes for CECS supported in refractory athletes; MTSS multifactorial — fascial traction one proposed mechanism

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. MTSS/CECS reviews. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Intermuscular Septa

ID: LL-FAS-02

Structure: Anterior and posterior crural intermuscular septa (+ transverse intermuscular septum)

Attachments: From crural fascia to fibula (anterior/posterior septa); transverse septum separates superficial/deep posterior

Continuity: Define four leg compartments with IO membrane and bones

Force Transmission: Compartmentalize pressure; guide neurovascular courses

Clinical Importance: Surgical landmarks for fasciotomy; septal tightness contributes to compartment pressures

Restrictions: Fibrosis after trauma/CECS

Assessment: Clinical compartment testing; intraoperative recognition

Treatment: Fasciotomy divides fascia/septa as needed for decompression

Evidence: Anatomic basis of four-compartment fasciotomy well established in trauma surgery

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Orthopaedic trauma compartment release literature.

RECORD: Compartment Fascia System (Integrated)

ID: LL-FAS-03

Structure: Integrated fascial compartment system of the leg

Attachments: Crural fascia + septa + IO membrane + tibial periosteum forming closed osteofascial spaces

Continuity: Knee fascia lata to ankle retinacula continuum

Force Transmission: Muscle expansion raises intracompartmental pressure; venous return assisted in normal compliance

Clinical Importance: Pathophysiology of acute and chronic compartment syndromes; MTSS periosteal-fascial interface

Restrictions: Reduced fascial compliance → CECS; trauma/bleed → acute CS

Assessment: Pain on passive stretch, firmness, neurovascular status; manometry protocols for CECS

Treatment: Acute = emergency fasciotomy. Chronic = conservative then fasciotomy

Evidence: Acute CS is surgical emergency — do not delay for imaging if clinically clear

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. ACS emergency guidelines themes.

7. Biomechanics

RECORD: Walking

ID: LL-BIO-01

Movement: Walking — lower leg contribution

Plane: Sagittal primary

Axis: Ankle mortise approximate

Primary Movers: TA eccentric heel strike; soleus midstance; gastroc late stance

Secondary Movers: TP controls pronation; fibularis control inversion

Stabilizers: Syndesmosis/ankle ligaments passive

Arthrokinematics: Tibial advancement over foot; fibula minor load share

Osteokinematics: Ankle DF then PF rockers

Joint Reaction Forces: Tibial axial load near BW to several x BW peak (model-dependent)

Torque: Plantarflexor moment late stance

Length-Tension: Soleus near mid-length midstance

EMG: Phasic TA/soleus/gastroc

Functional Activities: Community gait

Sport Applications: Base for run progressions

Compensations: Foot slap; early heel rise; vaulting

Common Dysfunctions: MTSS, Achilles, foot drop

Clinical Assessment: Observational gait; DF ROM; heel rise

Corrective Exercises: Heel-toe drills, calf capacity, cadence

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Running

ID: LL-BIO-02

Movement: Running — lower leg

Plane: Multiplanar

Axis: Ankle/foot complex

Primary Movers: Soleus/gastroc spring; TA swing/impact

Secondary Movers: TP arch; fibularis lateral stability

Stabilizers: Bone stress risk with load spikes

Arthrokinematics: Higher rates of loading than walking

Osteokinematics: Flight + stance; fore/mid/rearfoot strike variants

Joint Reaction Forces: Multiple x BW; tibial strain rises with speed/overstride

Torque: High Achilles/PF torque

Length-Tension: Tendon energy storage critical

EMG: High burst EMG

Functional Activities: Fitness/field running

Sport Applications: Endurance and team sports

Compensations: Overstride; crossover; excessive pronation or rigidity

Common Dysfunctions: MTSS, stress fracture, CECS, tendinopathy

Clinical Assessment: Video; step rate; shin pain map

Corrective Exercises: Cadence up, run-walk, calf capacity, surface progression

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Heel Strike

ID: LL-BIO-03

Movement: Heel strike / initial contact

Plane: Sagittal

Axis: Ankle

Primary Movers: TA eccentric

Secondary Movers: EHL/EDL assist

Stabilizers: Ankle mortise stability

Arthrokinematics: Rapid PF controlled by dorsiflexors

Osteokinematics: Heel rocker begins

Joint Reaction Forces: Impact transient

Torque: DF eccentric torque

Length-Tension: TA lengthening under load

EMG: High TA eccentric EMG

Functional Activities: Walking IC

Sport Applications: Rearfoot strike running

Compensations: Foot slap if weak TA

Common Dysfunctions: Anterior CS symptoms; TA tendinopathy rare

Clinical Assessment: Listen/watch foot slap; DF strength

Corrective Exercises: Eccentric TA lowers; heel walks

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Midstance

ID: LL-BIO-04

Movement: Midstance

Plane: Sagittal + frontal foot

Axis: Ankle/STJ

Primary Movers: Soleus eccentric tibial control; TP arch

Secondary Movers: Fibularis lateral

Stabilizers: Osteoligamentous ankle

Arthrokinematics: Tibia rotates forward over foot; STJ pronation then resupination begins

Osteokinematics: Ankle rocker

Joint Reaction Forces: High soleus demand

Torque: PF eccentric torque controlling DF

Length-Tension: Soleus continuous load

EMG: High soleus EMG

Functional Activities: Single-limb support

Sport Applications: Running midstance

Compensations: Excessive pronation or early heel rise

Common Dysfunctions: Achilles, soleus strain, PTTD, MTSS

Clinical Assessment: Single-leg stance; navicular drop clinical context

Corrective Exercises: Bent-knee heel raises; short foot; cadence

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Toe-Off

ID: LL-BIO-05

Movement: Toe-off / propulsion

Plane: Sagittal

Axis: Ankle/MTP

Primary Movers: Gastroc/soleus concentric; FHL/FDL toe flexion

Secondary Movers: TP/fibularis

Stabilizers: Windlass via plantar fascia

Arthrokinematics: Rapid PF; heel rise

Osteokinematics: Forefoot rocker

Joint Reaction Forces: Peak PF power walking

Torque: Large PF torque

Length-Tension: Achilles spring release in run

EMG: Peak gastroc/soleus EMG

Functional Activities: Push-off ADLs

Sport Applications: Sprint acceleration

Compensations: Reduced heel rise; hallux limitus compensation

Common Dysfunctions: Achilles tendinopathy; turf toe differential

Clinical Assessment: Heel-rise symmetry; push-off video

Corrective Exercises: Heel raises progressive; hallux mobility/strength

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Sprinting

ID: LL-BIO-06

Movement: Sprinting — lower leg demands

Plane: Sagittal dominant

Axis: Ankle

Primary Movers: Gastroc/soleus extreme PF power; TA swing recovery

Secondary Movers: FHL push-off

Stabilizers: Bone/soft-tissue high strain rates

Arthrokinematics: Forefoot dominance common; minimal contact times

Osteokinematics: Rapid stretch-shortening Achilles

Joint Reaction Forces: Very high tibial and Achilles loads

Torque: Maximal PF power

Length-Tension: SSC optimal stiffness

EMG: Maximal EMG bursts

Functional Activities: Track sprinting

Sport Applications: Speed sports

Compensations: Excess stiffness or overstride

Common Dysfunctions: Tennis leg; Achilles rupture risk factors; bone stress

Clinical Assessment: Sprint mechanics; calf capacity; prior injury history

Corrective Exercises: Graded exposures; heavy calf strength; technique

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

8. Pathologies

RECORD: Medial Tibial Stress Syndrome

ID: LL-PATH-01

Condition: Medial tibial stress syndrome (MTSS / shin splints)

Definition: Exercise-related pain along posteromedial tibial border from bone stress continuum / fascial traction — not a single histology

Mechanism: Repetitive loading exceeding tibial adaptation; traction of soleus/TP/fascial attachments proposed

Risk Factors: Novice runners, sudden volume increase, high BMI, female sex in some cohorts, prior MTSS, hard surfaces

Symptoms: Diffuse posteromedial shin pain with running that may linger after; tenderness along border

Red Flags: Focal pinpoint bony pain, night pain, hop pain severe — consider stress fracture; acute CS signs

Differential Diagnosis: Tibial stress fracture, CECS, saphenous neuropathy, soft-tissue strain

Clinical Examination: Palpation length of medial border; hop test; compartment symptoms screen

Special Tests: Clinical diagnosis primarily; hop/fulcrum for fracture suspicion

Imaging: MRI if stress fracture suspected or refractory; XR often normal early

Healing Timeline: Symptoms often 2-6+ weeks with load modification; bone adaptation longer

Conservative Management: Relative rest from impact, graded return, calf/hip strength, cadence increase, footwear/surface review

Surgical Management: Not indicated for MTSS proper

Rehabilitation Phases: Unload → cross-train → walk-run → build volume <10%/week heuristic

Return to Running: Pain-guided walk-run when daily pain settles

Return to Sport: When training loads match sport without rebound pain

Prognosis: Good with load management; recurrence if errors repeat

Clinical Guidelines: Sports medicine MTSS reviews; bone stress continuum framing

Evidence Level: Moderate — clinical consensus strong; mechanisms multifactorial

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty bone-stress/CECS/MTSS literature.

RECORD: Tibial Stress Fracture

ID: LL-PATH-02

Condition: Tibial stress fracture

Definition: Fatigue fracture of tibial cortex from repetitive overload on bone stress continuum

Mechanism: Accumulated microdamage > remodeling; common posteromedial cortex; anterior cortex high-risk tension side

Risk Factors: Endurance athletes, RED-S/low energy availability, female athlete triad spectrum, rapid mileage, prior BSI

Symptoms: Focal bony pain, night pain possible, pain with hopping; antalgic run

Red Flags: Anterior mid-tibial 'dreaded black line' high-risk; neurologic/vascular change; inability to WB

Differential Diagnosis: MTSS, CECS, tumor (rare), infection

Clinical Examination: Point tenderness, hop test, tuning fork limited value

Special Tests: Hop test; fulcrum

Imaging: MRI gold standard for early BSI; XR may lag 2-3 weeks; CT if delayed union

Healing Timeline: Low-risk often 6-8+ weeks; high-risk anterior much longer / may need surgery

Conservative Management: Unload (boot/crutches) per risk site; nutrition/energy availability; graded reload

Surgical Management: Selected high-risk anterior / nonunion may need fixation

Rehabilitation Phases: Protected WB → progressive loading → run-walk when pain-free criteria met

Return to Running: Only after clinical ± imaging clearance milestones

Return to Sport: Delayed until full training tolerance; site-dependent

Prognosis: Good for low-risk; guarded for anterior tension-side

Clinical Guidelines: Bone stress injury consensus themes (risk stratification)

Evidence Level: Strong imaging evidence; management risk-stratified

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty bone-stress/CECS/MTSS literature.

RECORD: Fibular Stress Fracture

ID: LL-PATH-03

Condition: Fibular stress fracture

Definition: Fatigue fracture of fibula from repetitive load

Mechanism: Repetitive running/jumping; distal fibula common in runners

Risk Factors: Endurance sports, training spikes, alignment factors

Symptoms: Lateral leg pain focal, activity-related

Red Flags: Same BSI red flags; common fibular nerve symptoms if proximal

Differential Diagnosis: Peroneal tendinopathy, lateral ankle instability, CECS lateral

Clinical Examination: Point tenderness along fibula; hop

Special Tests: Clinical + imaging

Imaging: MRI preferred; XR may be negative early

Healing Timeline: Often 6-8 weeks low-risk sites

Conservative Management: Load reduction, progressive return, address calf/evertors

Surgical Management: Rare

Rehabilitation Phases: Unload → cross-train → graded run

Return to Running: Pain-free criteria

Return to Sport: When lateral load tolerated

Prognosis: Generally favorable

Clinical Guidelines: BSI risk-stratification frameworks

Evidence Level: Moderate

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty bone-stress/CECS/MTSS literature.

RECORD: Chronic Exertional Compartment Syndrome

ID: LL-PATH-04

Condition: Chronic exertional compartment syndrome (CECS)

Definition: Reversible elevated intracompartmental pressure with exercise causing pain/neuro symptoms that resolve with rest

Mechanism: Fascial noncompliance + muscle volume expansion with exertion

Risk Factors: Running athletes, anterior and deep posterior most common, prior MTSS misdiagnosis

Symptoms: Predictable onset at same time/distance; tightness, weakness, paresthesia; resolves after stopping

Red Flags: Pain at rest, severe neuro deficit, or trauma history — think acute CS emergency

Differential Diagnosis: MTSS, stress fracture, nerve entrapment, POPA, arterial endofibrosis (rare cyclists)

Clinical Examination: Post-exercise firmness, neuro exam, reproduction with sport-specific exertion

Special Tests: Invasive manometry protocols still reference standard; some centers use continuous/other methods

Imaging: Manometry diagnostic aid; MRI/SPECT selected adjuncts — clinical correlation essential

Healing Timeline: Symptoms chronic until load adapted or fasciotomy

Conservative Management: Gait retraining (forefoot/cadence cues evolving evidence), activity mod, fascial mobility adjunct

Surgical Management: Fasciotomy for refractory confirmed CECS — discuss risks (hernia, recurrence, nerve)

Rehabilitation Phases: Conservative trial → post-fasciotomy progressive loading if surgery

Return to Running: Graded after symptoms controlled / post-op milestones

Return to Sport: Often good after successful fasciotomy in selected athletes

Prognosis: Variable; surgery helps many refractory cases

Clinical Guidelines: CECS clinical reviews; acute vs chronic distinction mandatory

Evidence Level: Moderate diagnostic heterogeneity; surgery outcomes generally favorable in series

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty bone-stress/CECS/MTSS literature.

RECORD: Gastrocnemius Tear

ID: LL-PATH-05

Condition: Gastrocnemius tear (tennis leg — typically medial head)

Definition: Myofascial junction strain/tear of gastrocnemius, usually medial head

Mechanism: Sudden push-off or eccentric DF with extended knee (tennis, sprint)

Risk Factors: Middle-aged athletes, inadequate warm-up, prior calf injury

Symptoms: Sudden pop/stabbing calf pain, bruising, limp; Thompson negative if Achilles intact

Red Flags: Positive Thompson / gap at Achilles = rupture; massive swelling + dyspnea = PE/DVT workup pathway

Differential Diagnosis: Achilles rupture, plantaris rupture, soleus strain, DVT

Clinical Examination: Palpation medial gastroc MJ, strength PF knee extended vs flexed, Thompson

Special Tests: Thompson test to exclude Achilles

Imaging: US excellent for tear localization; MRI if unclear

Healing Timeline: Grade-dependent 2-8+ weeks typical functional recovery window

Conservative Management: PEACE&LOVE/POLICE principles, early protected motion, progressive loading — avoid prolonged immobilization

Surgical Management: Rare for isolated MJ tear

Rehabilitation Phases: Protect → isometric → concentric → eccentric → plyometric/sprint

Return to Running: When calf capacity and pain allow walk-run

Return to Sport: Criteria-based sprint/cut return

Prognosis: Good; reinjury risk if rushed

Clinical Guidelines: Muscle strain rehab evidence themes

Evidence Level: Good clinical consensus; US useful

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty bone-stress/CECS/MTSS literature.

RECORD: Soleus Strain

ID: LL-PATH-06

Condition: Soleus strain

Definition: Strain of soleus muscle — often deep calf pain, may be subacute

Mechanism: Repetitive eccentric load midstance or sudden overload

Risk Factors: Runners, hill training, inadequate calf capacity

Symptoms: Deep calf pain worse with bent-knee heel raise; less dramatic than tennis leg

Red Flags: DVT signs; Achilles rupture signs

Differential Diagnosis: DVT, MTSS, CECS, radiculopathy

Clinical Examination: Pain with knee-flexed heel raise; deep palpation

Special Tests: Clinical; Thompson negative

Imaging: US/MRI if needed to confirm and grade

Healing Timeline: Often 3-8 weeks

Conservative Management: Load modification, bent-knee heel raise progressive heavy slow

Surgical Management: Not typical

Rehabilitation Phases: Isometrics → HSR soleus → run-walk

Return to Running: When soleus loading tolerated

Return to Sport: When sport volumes tolerated

Prognosis: Good with progressive loading

Clinical Guidelines: Calf strain rehab principles

Evidence Level: Moderate — under-recognized clinically

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty bone-stress/CECS/MTSS literature.

RECORD: Peroneal Tendinopathy

ID: LL-PATH-07

Condition: Peroneal (fibularis) tendinopathy

Definition: Overuse pain/dysfunction of fibularis longus/brevis tendons at retromalleolar / cuboid regions

Mechanism: Repetitive eversion load, ankle instability, training errors, cavovarus foot sometimes

Risk Factors: Runners, dancers, lateral ankle sprain history, unstable retinaculum

Symptoms: Lateral ankle/foot pain with eversion/push-off; swelling behind lateral malleolus

Red Flags: Acute dislocation of tendons; fracture of 5th MT; infection

Differential Diagnosis: Lateral ankle sprain, sural nerve, 5th MT stress, cuboid syndrome

Clinical Examination: Palpation tendons, resisted eversion, subluxation on DF-eversion

Special Tests: Resisted eversion; dynamic subluxation observation

Imaging: US first-line dynamic; MRI for tears

Healing Timeline: Tendinopathy months; tears may need longer/surgery

Conservative Management: Load management, progressive eversion loading, ankle stability, orthoses selected

Surgical Management: Repair/groove deepening/retinaculum for subluxation or selected tears

Rehabilitation Phases: Isometrics → isotonic eversion → plyometric/uneven surface

Return to Running: When eversion capacity and pain allow

Return to Sport: When cutting on uneven ground tolerated

Prognosis: Good if instability addressed

Clinical Guidelines: Peroneal tendon clinical reviews

Evidence Level: Moderate evidence for progressive loading + stability

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty bone-stress/CECS/MTSS literature.

9. Rehabilitation Pathways

RECORD: MTSS Rehabilitation

ID: LL-REHAB-01

Condition: Medial tibial stress syndrome

Rehabilitation Phase: Load modification to graded return

Goals: Settle medial tibial pain; restore run capacity without bone-stress progression

Pain Management: Relative rest from aggravating impact; ice after activity optional; avoid NSAID overuse masking bone pain

Weight Bearing Status: WBAT; reduce running first rather than full NWB unless fracture suspected

ROM Goals: Ankle DF mobility if limited; hip mobility as needed

Strength Goals: Soleus/gastroc, TP, hip abductors progressive

Motor Control: Cadence increase (~5-10%), reduce overstride

Balance Training: Single-leg stance progressions

Proprioception: Uneven surface graded later

Functional Training: Bike/swim/elliptical cross-train

Sport-Specific Progression: Walk-run schedule with pain rules (e.g. pain $\leq 2-3/10$ and not rising day-to-day)

Criteria to Progress: Palpation less irritable; daily pain settled; hop comfortable before run build

Return to Running: Walk-run when symptoms allow — monitor 24h response

Return to Sport: When volume/intensity match sport without delayed pain

Outcome Measures: NPRS, running volume log, single-leg hop

Clinical Guidelines: MTSS load-management reviews; exclude BSI if red flags

Evidence: Load modification cornerstone; evidence for specific modalities mixed

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Tibial Stress Fracture Rehabilitation

ID: LL-REHAB-02

Condition: Tibial stress fracture

Rehabilitation Phase: Protected healing then reload (risk-site dependent)

Goals: Bone healing; safe graded impact return; correct energy availability

Pain Management: Unload until pain-free walking; nutrition referral if RED-S suspected

Weight Bearing Status: Boot/crutches per site/risk — high-risk anterior more protected

ROM Goals: Maintain ankle/knee/hip mobility in boot constraints

Strength Goals: Cross-training strength; progressive calf/hip when allowed

Motor Control: Gait retraining before run

Balance Training: When WB allows

Proprioception: Progressive

Functional Training: Deep-water run, bike per pain

Sport-Specific Progression: Walk → walk-run → continuous → speed/hills last

Criteria to Progress: Pain-free ADLs; clinician ± imaging milestones for high-risk

Return to Running: Only after cleared — often 6-8+ weeks low-risk

Return to Sport: Site-dependent; high-risk delayed substantially

Outcome Measures: Pain, hop, training log, RED-S screens

Clinical Guidelines: BSI risk stratification consensus themes

Evidence: MRI grading informs prognosis; high-risk sites need caution

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. BSI consensus literature.

RECORD: CECS Rehabilitation

ID: LL-REHAB-03

Condition: Chronic exertional compartment syndrome

Rehabilitation Phase: Conservative trial ± post-fasciotomy

Goals: Reduce exertional pressure symptoms; return to desired sport

Pain Management: Stop at symptom onset early in rehab; avoid pushing through neuro symptoms

Weight Bearing Status: Full

ROM Goals: Ankle mobility; fascial glide adjunct

Strength Goals: Capacity of involved compartment muscles without symptom spike

Motor Control: Gait/run form cues (cadence, strike) — emerging supportive evidence

Balance Training: Standard ankle-leg progressions

Proprioception: Progressive

Functional Training: Cross-train below symptom threshold

Sport-Specific Progression: Timed run intervals below onset threshold then extend

Criteria to Progress: Delayed symptom onset; acceptable for goals OR proceed to surgery discussion

Return to Running: Symptom-threshold based

Return to Sport: Post-fasciotomy typically months with graded loading

Outcome Measures: Time-to-symptom onset, VAS, sport participation

Clinical Guidelines: Confirm diagnosis before fasciotomy; distinguish acute CS

Evidence: Fasciotomy success rates good in selected series; conservative first reasonable

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Gastrocnemius Tear Rehabilitation

ID: LL-REHAB-04

Condition: Gastrocnemius tear (tennis leg)

Rehabilitation Phase: Acute protect to RTS

Goals: Restore calf capacity and sprint/cut confidence

Pain Management: Early relative rest, compression, elevate; crutches if antalgic

Weight Bearing Status: WBAT as pain allows; avoid prolonged cast

ROM Goals: Early gentle DF within pain; restore full DF

Strength Goals: Isometrics → concentric heel raises → eccentric → loaded

Motor Control: Heel-toe gait restoration

Balance Training: Bilateral → unilateral

Proprioception: Progressive

Functional Training: Bike early; pool

Sport-Specific Progression: Jog → stride → sprint → sport COD

Criteria to Progress: Pain-free heel raises capacity, hop, sprint without apprehension

Return to Running: When walking and calf strength adequate

Return to Sport: Often 4-8+ weeks grade-dependent

Outcome Measures: Heel-rise endurance, NPRS, hop

Clinical Guidelines: Exclude Achilles rupture and DVT

Evidence: Early loading better than prolonged immobilization for muscle strain

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Soleus Strain Rehabilitation

ID: LL-REHAB-05

Condition: Soleus strain

Rehabilitation Phase: Progressive soleus loading

Goals: Restore bent-knee calf capacity for running midstance demands

Pain Management: Reduce hills/speed initially

Weight Bearing Status: Full

ROM Goals: Ankle DF with knee flexed

Strength Goals: Seated/bent-knee heavy slow heel raises primary

Motor Control: Midstance control drills

Balance Training: Single-leg

Proprioception: Progressive

Functional Training: Cross-train

Sport-Specific Progression: Walk-run → hills last

Criteria to Progress: Pain-free soleus loading volume

Return to Running: When bent-knee raises and walk pain-free

Return to Sport: When run volume tolerated

Outcome Measures: Bent-knee heel-rise endurance, run log

Clinical Guidelines: Differentiate DVT/MTSS

Evidence: Muscle loading principles; soleus-specific bias important

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Peroneal Tendinopathy Rehabilitation

ID: LL-REHAB-06

Condition: Peroneal tendinopathy

Rehabilitation Phase: Progressive tendon loading + stability

Goals: Improve eversion load tolerance; control subluxation risk

Pain Management: Reduce provocative cutting/uneven volume; isometrics for analgesia

Weight Bearing Status: Full

ROM Goals: Ankle inversion/eversion mobility without forcing unstable tendons

Strength Goals: Isometric eversion → isotonic → functional

Motor Control: Single-leg landing with frontal control

Balance Training: Uneven surface graded

Proprioception: Ankle disc/star excursion

Functional Training: Lateral band walks, side steps

Sport-Specific Progression: Linear → lateral → cutting

Criteria to Progress: Pain rules + strength + no subluxation apprehension

Return to Running: When daily pain settled with eversion loading

Return to Sport: When COD on uneven ground tolerated

Outcome Measures: NPRS, eversion strength, FAAM

Clinical Guidelines: Assess retinaculum/subluxation — surgical if mechanical instability

Evidence: Progressive loading + neuromuscular control supported clinically

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Fibular Stress Fracture Rehabilitation

ID: LL-REHAB-07

Condition: Fibular stress fracture

Rehabilitation Phase: Unload to graded impact

Goals: Bone healing; return to run without recurrence

Pain Management: Reduce impact until pain-free ADLs

Weight Bearing Status: Modify per pain; boot selected cases

ROM Goals: Maintain ankle mobility

Strength Goals: Hip/calf/eversion progressive when allowed

Motor Control: Gait retraining

Balance Training: When WB comfortable

Proprioception: Progressive

Functional Training: Cross-train

Sport-Specific Progression: Walk-run build

Criteria to Progress: Pain-free hop and ADLs

Return to Running: After clinical clearance window (~6-8 weeks typical low-risk)

Return to Sport: When training loads restored

Outcome Measures: Pain, hop, training log

Clinical Guidelines: BSI risk stratification

Evidence: Similar principles to other low-risk BSI

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

10. Evidence and Guidelines

RECORD: Key Evidence and Guidelines — Lower Leg Module

ID: LL-EVID-01

Topic: Foundational references for Kinora lower leg AI orientation

Anatomy: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH.

Biomechanics: Neumann DA. Kinesiology of the Musculoskeletal System.

Assessment: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

Compartment: Acute compartment syndrome is a surgical emergency. CECS is distinct — exertional, reversible with rest; manometry protocols used for confirmation.

Bone Stress: MRI preferred for early BSI; risk-stratify anterior tibial cortex as higher risk; screen energy availability/RED-S when recurrent.

MTSS: Clinical diagnosis; load management first; exclude stress fracture if focal/night/hop pain.

Calf: Always differentiate tennis leg vs Achilles rupture (Thompson) vs DVT.

AI Use Note: Timelines and pressure cutoffs vary by protocol — prefer current orthopaedic/sports medicine guidance and individual assessment.

References: See fields above.